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Course Welcome

Module 1 Introduction

Lesson 1: Logistic Regression Part I

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Module 1 Summative Assessment

Graded Assignment: Module 1 Summative Assessment

Submitted

Module 1 Summary

# Module 1 Summative Assessment

## Review Learning Objectives

ⓘ

You ran out of attempts. If you'd like to try again, you will have attempts available in 7h 59m

### Assignment details

| Due                  | Attempts                          |
|----------------------|-----------------------------------|
| Jan 21, 11:59 PM IST | 0 left (1 attempts every 8 hours) |
| Submitted            | Jan 23, 12:10 PM IST              |

Your grade

To pass you need at least 80%. We keep your highest score.

100%

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

### Instructions

1.

Which of the following statement(s) is/are true about a binary logistic regression?

1 / 1 point
- (I) The predictor variables ( $x$ ) are continuous, with values between 0 and

(II) The response variable ( $y$ ) is continuous, with values between 0 and 1.

(III) The outputs of the sigmoid function are continuous, with values between 0 and 1.

☐ I only

☐ II only

☒ III only

☐ II and III only

☐ I, II, and III

✔ Correct

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2.

Let  $Y$  denote whether a student get admitted to attend Illinois Tech, with  $Y = 1$  as successfully being admitted, and  $Y = 0$ , otherwise. Also, let

1 / 1 point
- $X_1$  denote the math scores (0-100)
  - $X_2$  denote the reading scores (0-100)

of a particular standardize exam that a student need to take to gain admission. We collected n-data and run a logistic regression to obtain

$$\ln\left(\frac{\pi}{1-\pi}\right) = 0.13 + 0.05X_1 + 0.04X_2$$

where

$$\pi = \Pr(Y = 1|X_1 = x_1, X_2 = x_2).$$

Suppose a student has a math scores of 78% and the reading scores of 82%. If the student retake the math test and scores 83%. Compute the following change of odds in being admitted to Illinois Tech.

$$\frac{\text{Odds}(x_0 + 1) - \text{Odds}(x_0)}{\text{Odds}(x_0)} \times 100\%$$

- ☒ 5.127%

☐ 13.883%

☐ 4.081%

☐ 0.05%

☐ 0.04%
- ✔ Correct

3.

Suppose we want to model the profitability of a movie ( $y$ ) based on the opening-weekend revenue ( $x$ , in millions of dollar). We let  $y = 1$  if the movie is profitable, that is, the revenue is larger than the budget, and not profitable as  $y = 0$ . Based on the data from 100 movies in the past 50 years, we obtain

1 / 1 point

$$\log(\text{odds}) = -2.27 + 1.276x$$

Which of the following is/are correct?

- (I) The odds ratio is given by  $e^{1.276}$ .

(II) For a one-unit increase in  $x$ , we expect a 258.2% increase in the odds.

(III) For a one-unit increase in  $x$ , we expect the odds that the movie will be profitable increase by a factor of 3.58.

☐ I only

☐ II only

☐ III only

☐ II and III only

☒ I, II, and III

✔ Correct

4.

Suppose that the probability of an event is 0.2, what is the odds of the event occurring?

1 / 1 point
- ☐ 0.2

☐ 0.8

☐ 0.12

☒ 0.25

☐ 0.3
- ✔ Correct

